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Why This Problem Matters

Industrial manipulators are rarely operated under one fixed model.

- Payload changes alter inertia and gravity loading.
- Friction drift and disturbances break fixed-gain assumptions.
- Direct torque-space RL can fail during exploration.

Research question

Can an interpretable fuzzy supervisor make residual RL safer and more useful under plant mismatch?

Key message

The goal is not to replace classical control everywhere. The goal is to add adaptive residual authority only where mismatch makes it valuable.

Positioning the Idea

What fuzzy control gives

- Linguistic, interpretable rule base.
- Immediate stabilizing baseline.
- Low sample cost and predictable response.

What residual RL gives

- Learned compensation for unmodeled shifts.
- Continuous torque correction.
- Adaptation through reward feedback.

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{fuzzy}} + g(\mathbf{e}, \dot{\mathbf{e}}, \xi)\boldsymbol{\tau}_{\text{RL}}$$

Controller Architecture

Hybrid control law

$$\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{fuzzy}} + g(\boldsymbol{e}, \dot{\boldsymbol{e}}, \xi)\boldsymbol{\tau}_{\text{RL}}, \quad \|\boldsymbol{\tau}_{\text{RL}}\|_\infty \le \bar{\tau}_r.$$

- Mamdani fuzzy supervisor uses e and $\dot{e}$.
- TD3 or SAC learns bounded residual torque.
- Gate $g \in [0, 1]$ suppresses residual action during nominal tracking.

Uncertainty gate

$$g = \mathrm{clip}\left(\left(\frac{\|\boldsymbol{e}\|}{\epsilon_g}\right)^2 + 0.2\left(\frac{\xi}{\epsilon_g}\right)^2 + 0.05\eta, 0, 1\right).$$

ξ is persistent tracking error; η is normalized velocity-error activity.

What Changed From the Original Claim

Old inflated claim

Hybrid Fuzzy-TD3 is universally superior for trajectory tracking.

Revised defensible claim

Fuzzy-supervised residual RL is a robustness layer: it improves shifted-plant tracking and avoids direct-RL failures, while uncertainty gating is needed to preserve nominal performance.

- No DQN comparison is claimed.
- Comparisons are against PID, fuzzy-only, model-based baselines, TD3, SAC, DDPG, residual TD3, and residual SAC.
- 6-DOF validation is stated as required future work, not as completed evidence.

Experimental Protocol

Benchmark

- Two-link rigid-body manipulator.
- Step size: 20 ms; horizon: 6 s.
- Torque saturation: 35 Nm.
- Disturbances begin at 2.5 s.

Scenarios

- Nominal plant.
- Payload doubled, friction increased.
- Severe shift: payload tripled, stronger friction and sinusoidal disturbance.

Five seeds, 12000 training steps per learned controller, and 20 evaluation rollouts per controller per scenario.

Baselines and Metrics

Baselines

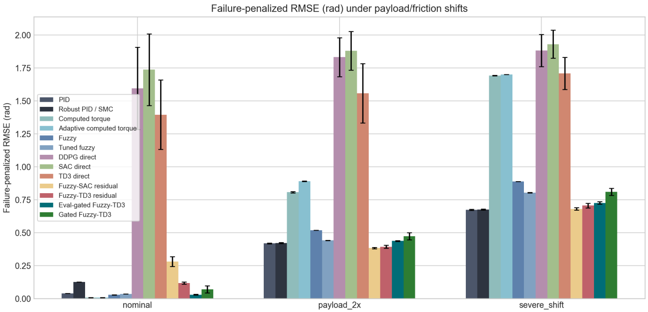
- PID and robust PID/sliding-mode term.
- Computed torque and adaptive computed torque.
- Fuzzy-only and tuned fuzzy.
- Direct TD3, SAC, DDPG.
- Residual TD3, residual SAC, gated Fuzzy-TD3.

Metrics

- Joint RMSE.
- Failure-penalized RMSE.
- Safety-violation rate.
- Torque smoothness RMS.
- Completion rate.

Failure-penalized RMSE prevents early-failing direct-RL rollouts from looking deceptively good.

Main Result: Shifted-Plant Robustness



Interpretation

- Model-based control dominates nominal tracking.
- Direct RL fails frequently.
- Residual SAC and residual TD3 improve under payload/friction shifts.
- Gating nearly fixes nominal degradation.

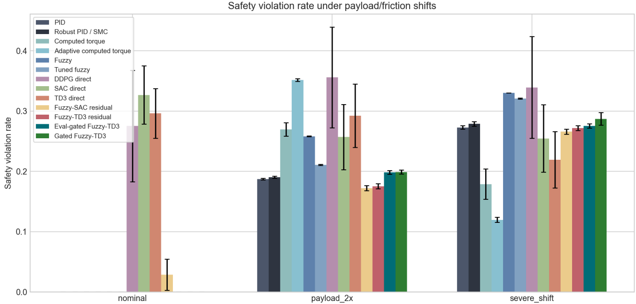
Numerical Takeaways

Scenario	Controller	Penalized RMSE	Safety	Completion
Nominal	Fuzzy	0.0284	0.000	1.00
Nominal	Eval-gated Fuzzy-TD3	0.0299	0.000	1.00
Payload-2x	Fuzzy-SAC residual	0.3839	0.172	1.00
Payload-2x	Fuzzy-TD3 residual	0.3936	0.175	1.00
Payload-2x	PID	0.4190	0.187	1.00
Severe shift	PID	0.6739	0.273	1.00
Severe shift	Fuzzy-SAC residual	0.6804	0.266	1.00
Severe shift	Fuzzy-TD3 residual	0.7065	0.272	1.00

Key message

Residual SAC is the strongest learned controller; residual TD3 remains competitive; both avoid the failure behavior of direct RL.

Safety: Direct RL vs Supervised Residual RL

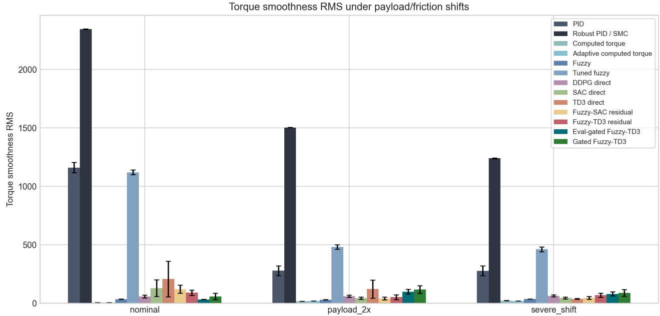


What the plot shows

- Direct TD3/SAC/DDPG often violate safety thresholds.
- Residual policies inherit fuzzy baseline structure.
- Bounded residual authority limits worst-case exploration.

This is empirical safety, not a formal proof.

Torque Smoothness



Why this matters

- PID and robust PID can track shifts but may be torque-rough.
- Residual SAC is close to PID tracking in severe shift with much smoother torque.
- Smoothness is important for actuator wear and deployment credibility.

Gating Fixes the Main Weakness

Problem

Always-active residual TD3 hurts nominal tracking:

$$0.1184 \text{ rad}$$

versus fuzzy-only:

$$0.0284 \text{ rad}.$$

Gated result

Evaluation-gated Fuzzy-TD3:

$$0.0299 \text{ rad}$$

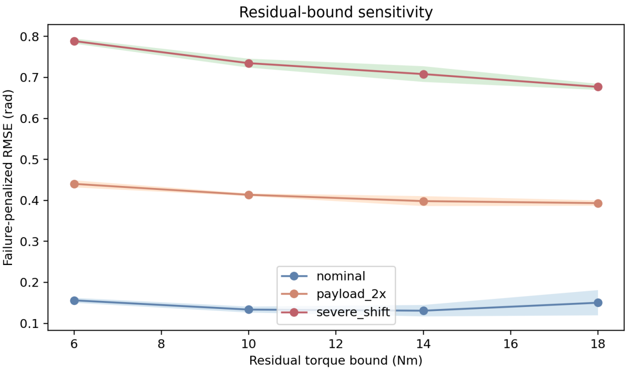
with mean nominal gate:

$$g = 0.034.$$

Key message

The publishable direction is uncertainty-gated residual learning, not always-on residual action.

Ablation: Residual Authority



Trade-off

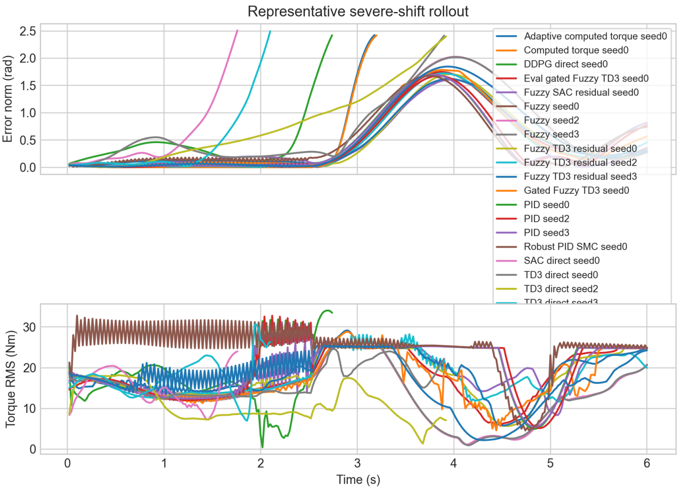
- Larger residual bounds improve shifted-plant tracking.
- Every nonzero bound hurts nominal tracking.
- This directly motivates uncertainty gating.

Ablation Table

Scenario	0 Nm	6 Nm	10 Nm	14 Nm	18 Nm
Nominal	0.0292	0.1561	0.1335	0.1308	0.1502
Payload-2x	0.5192	0.4403	0.4136	0.3982	0.3933
Severe shift	0.8891	0.7887	0.7348	0.7081	0.6769

Values are failure-penalized RMSE means over five seeds and twenty evaluation rollouts per setting.

Representative Severe-Shift Rollout



Severe-shift traces show where residual learning helps and where classical baselines remain competitive.

Safety and Stability Framing

What we can honestly claim

$$\|\boldsymbol{\tau}_{\mathrm{RL}}\|_\infty \leq \bar{\tau}_r, \quad 0 \leq g \leq 1$$

so residual authority is explicitly bounded.

- Learned residual enters the controller as a bounded additive torque.
- Passivity-style filtering can limit residual work against tracking-error velocity.
- This supports practical boundedness under standard assumptions.

Key message

This is not a complete Lyapunov theorem for the neural policy. A journal version should add CBF, Lyapunov-constrained RL, or passivity-constrained policy optimization.

Limitations

- Current validated benchmark is 2-DOF, not yet UR5/Panda/KUKA-scale.
- PyBullet installation failed locally because Microsoft C++ Build Tools were missing.
- The gate is hand-designed/evaluation-time, not yet learned or formally verified.
- Longer training budgets and hardware-in-the-loop latency tests are still needed.

Journal-readiness status

Strong preliminary manuscript and reproducible benchmark; not yet a final high-impact journal submission until 6-DOF or hardware validation is completed.

Next Experiments Before Submission

1. Run the same protocol on UR5, Franka Panda, KUKA iiwa, or xArm6 in PyBullet/MuJoCo/Gazebo.
2. Replace hand-designed gate with an uncertainty estimator or learned gate.
3. Add CBF or Lyapunov-style residual safety filter.
4. Extend to 10 seeds and longer training horizon.
5. Report hardware or real-time edge latency if possible.

Key message

The upgraded research idea is credible because the experiments now reveal both the benefit and the boundary of the method.

Final Takeaway

Fuzzy-supervised residual RL is best framed as

an uncertainty-triggered adaptation layer

rather than a universal controller replacement.

Most defensible result

Residual SAC/TD3 improve robustness under payload and friction shift, direct RL fails frequently, and gating preserves nominal fuzzy-control performance.

Thank you